

Evidence-selection counts

Prenatal chronic noise exposure in rodents meta-analysis

Working document | 28 August 2026

Scope: Identification and screening counts use records/reports; the final inclusion count uses studies represented in the meta-analysis dataset.

Records identified by database or search source

Database or search source	Records
Scopus	1,404
Web of Science	1,842
PubMed	1,056
PsycINFO	408
OpenAlex	681
Google Scholar - supplementary search	489
BASE	9
Total records identified	5,889

Check: $1,404 + 1,842 + 1,056 + 408 + 681 + 489 + 9 = 5,889$.

Deduplication, screening, and final inclusion

Selection stage	Records	Reconciliation note
Records identified	5,889	Sum across all database and search-source results
Duplicates removed before screening	3,576	Combined deduplication, including repeated BASE records
Unique records screened	2,313	$5,889 - 3,576$
Title-and-abstract exclusions	2,226	Reasons are not reported at this stage
Full-text reports sought from formal search	87	$2,313 - 2,226$
Additional reports identified through other methods	4	Three benchmark/general-search papers and one related report

Selection stage	Records	Reconciliation note
Total full-text reports sought	91	87 + 4
Full-text reports not retrieved	1	Not included among assessed reports
Full-text reports assessed	90	91 - 1
Full-text reports excluded	74	Primary reasons shown below
Included through Rayyan	12	Formal search pathway
Additional included reports	4	Other-method pathway
Studies included in the meta-analysis	16	90 - 74

Full-text exclusion reasons

Primary exclusion reason	Reports
Wrong exposure	60
Wrong outcome	4
Wrong publication type	4
Wrong population	4
Foreign language	2
Total full-text exclusions	74

Foreign language received priority when a record had multiple active Rayyan labels. The one report that could not be retrieved is reported separately from the 74 assessed full-text exclusions.

Arithmetic reconciliation

1. $5,889 - 3,576 = 2,313$ unique records screened
2. $2,313 - 2,226 = 87$ full-text reports sought from the formal search
3. $87 + 4 = 91$ total full-text reports sought
4. $91 - 1 = 90$ full-text reports assessed
5. $90 - 74 = 16$ studies included in the meta-analysis

Source files

- Title-and-abstract decisions: PCN_screening/title_abstract_2026-08-28_23-45-55/articles.csv
- Full-text decisions: PCN_screening/full_text_2026-08-28_23-47-26/articles.csv
- Additional benchmark/general-search papers: PCN_screening/Benchmark/

- Final included-study dataset: Data/PCN_data_ext_checking_v.3.csv

Appendix A. Reports excluded after full-text assessment (n = 74)

Each report has one primary exclusion reason. Foreign language received priority when a report had multiple active labels. The unavailable report was not assessed and appears in Appendix B.

Rayyan ID	Reference	Primary reason
386283305	Perinatal stress and development : Anxious-like behavior in the rat. Stress perinatal et developpement emotionnel chez le rat: Implications pour l'etude du developpement humain. (2001). DOI: https://dx.doi.org/10.3917/enf.533.0293 .	Foreign language
386283784	The effects of prenatal sound stress on the spatial learning and memory of rat's male offspring.	Foreign language
386282493	Behavioral despair in mice after prenatal stress. JOURNAL OF PHYSIOLOGY AND BIOCHEMISTRY. (2000). DOI: 10.1007/BF03179902.	Wrong exposure
386282504	Effects of prenatal stress on anxiety-like behavior and nociceptive response in rats. JOURNAL OF INTEGRATIVE NEUROSCIENCE. (2015). DOI: 10.1142/S0219635215500107.	Wrong exposure
386282511	Vulnerability and resilience to prenatal stress exposure: behavioral and molecular characterization in adolescent rats. TRANSLATIONAL PSYCHIATRY. (2023). DOI: 10.1038/s41398-023-02653-6.	Wrong exposure
386282570	Prenatal Stress Alters the Development of Socioemotional Behavior and Amygdala Neuron Excitability in Rats. NEUROPSYCHOPHARMACOLOGY. (2015). DOI: 10.1038/npp.2015.55.	Wrong exposure
386282672	Prenatal Stress Enhances Excitatory Synaptic Transmission and Impairs Long-Term Potentiation in the Frontal Cortex of Adult Offspring Rats. PLOS ONE. (2015). DOI: 10.1371/journal.pone.0119407.	Wrong exposure
386282675	Fluoxetine normalizes the effects of prenatal maternal stress on depression- and anxiety-like behaviors in mouse dams and male offspring. BEHAVIOURAL BRAIN RESEARCH. (2016). DOI: 10.1016/j.bbr.2016.05.062.	Wrong exposure
386282676	Moderation of the transgenerational transference of antenatal stress-induced anxiety. TRANSLATIONAL PSYCHIATRY. (2021). DOI: 10.1038/s41398-021-01383-x.	Wrong exposure
386282678	The Modification of Offspring Stress-Related Behavior and the Expression of Drd1, Drd2, and Nr3c1 by a Western-Pattern Diet in Mus Musculus. INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES. (2022). DOI: 10.3390/ijms23169245.	Wrong exposure
386282708	Testosterone is involved in mediating the effects of prenatal stress in male guinea pig offspring. JOURNAL OF PHYSIOLOGY-LONDON. (2011). DOI: 10.1113/jphysiol.2010.200543.	Wrong exposure
386282729	Prenatal noise stress impairs HPA axis and cognitive performance in mice. SCIENTIFIC REPORTS. (2017). DOI: 10.1038/s41598-017-09799-6.	Wrong exposure
386282742	Hyperexcitability of VTA dopaminergic neurons in male offspring exposed to physical or psychological prenatal stress. PROGRESS IN NEURO-PSYCHOPHARMACOLOGY & BIOLOGICAL PSYCHIATRY. (2020). DOI: https://dx.doi.org/10.1016/j.pnpbp.2020.109923 .	Wrong exposure
386282745	Prenatal stress produces anxiety- and depression-related behavior particularly in male Sprague-Dawley rats. INTERNATIONAL JOURNAL OF NEUROPSYCHOPHARMACOLOGY. (2006).	Wrong exposure

Rayyan ID	Reference	Primary reason
386282763	Prenatal Stress Modulates Placental and Fetal Serotonin Levels and Determines Behavior Patterns in Offspring of Mice. INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES. (2024). DOI: 10.3390/ijms252413565.	Wrong exposure
386282782	Antidepressant and anxiolytic efficacy of single, chronic and concomitant use of vortioxetine, dapoxetine and fluoxetine in prenatally stressed rats. DOI: 10.21307/ane-2019-002.	Wrong exposure
386282801	Open field behavior of C57BL/6J mice as a function of age, experience, and prenatal maternal stress. PSYCHONOMIC SCIENCE. (1964). DOI: https://dx.doi.org/10.3758/BF03342972 .	Wrong exposure
386282810	Action of antidepressant drugs on maternal stress-induced hypoactivity in female rats. METHODS AND FINDINGS IN EXPERIMENTAL AND CLINICAL PHARMACOLOGY. (1999). DOI: 10.1358/mf.1999.21.4.538181.	Wrong exposure
386282812	INVOLVEMENT OF PGLUR1, EAAT2 AND EAAT3 IN OFFSPRING DEPRESSION INDUCED BY PRENATAL STRESS. NEUROSCIENCE. (2013). DOI: 10.1016/j.neuroscience.2013.04.031.	Wrong exposure
386282821	Duloxetine prevents the effects of prenatal stress on depressive-like and anxiety-like behavior and hippocampal expression of pro-inflammatory cytokines in adult male offspring rats. INTERNATIONAL JOURNAL OF DEVELOPMENTAL NEUROSCIENCE. (2016). DOI: 10.1016/j.ijdevneu.2016.09.005.	Wrong exposure
386282858	[Characterization of depression-like behavior in prenatally stressed female rats during ovary cycle]. Rossiiskii fiziologicheskii zhurnal imeni I.M. Sechenova. (2012).	Wrong exposure
386282859	The impact of mitochondrial aldehyde dehydrogenase (ALDH2) activation by Alda-1 on the behavioral and biochemical disturbances in animal model of depression. BRAIN BEHAVIOR AND IMMUNITY. (2016). DOI: 10.1016/j.bbi.2015.08.004.	Wrong exposure
386282868	Prenatal stress alters sensitivity to benzodiazepines in adult rats. NEUROSCIENCE LETTERS. (2015). DOI: 10.1016/j.neulet.2015.02.015.	Wrong exposure
386282921	Ferulic Acid Improves Depressive-Like Behavior in Prenatally-Stressed Offspring Rats via Anti-Inflammatory Activity and HPA Axis. INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES. (2019). DOI: 10.3390/ijms20030493.	Wrong exposure
386283005	Sex-Dependent Effects of Prenatal Stress on Learned Helplessness and Anxiety-Related Behaviours in Wistar Rats. DOI: 10.4236/jbbs.2015.57026.	Wrong exposure
386283023	Sex- and age- dependent effect of pre-gestational chronic stress and mirtazapine treatment on neurobehavioral development of Wistar rat offspring. PLOS ONE. (2022). DOI: 10.1371/journal.pone.0255546.	Wrong exposure
386283046	Prenatal maternal stress alters depression-related symptoms in a strain- and sex-dependent manner in rodent offspring. LIFE SCIENCES. (2020). DOI: 10.1016/j.lfs.2020.117597.	Wrong exposure
386283175	Prenatal stress induces a depressive-like phenotype in adolescent rats: The key role of TGF- β 1 pathway. FRONTIERS IN PHARMACOLOGY. (2022). DOI: 10.3389/fphar.2022.1075746.	Wrong exposure
386283209	CRTC1 signaling involvement in depression-like behavior of prenatally stressed offspring rat. BEHAVIOURAL BRAIN RESEARCH. (2021). DOI: 10.1016/j.bbr.2020.113000.	Wrong exposure
386283223	[The study of the behavioral reactions and duration of ultrasonic vocalization in rats from females stressed during pregnancy]. Zhurnal neurologii i psikiatrii imeni S.S. Korsakova. (2025). DOI: 10.17116/jnevro2025125031117.	Wrong exposure

Rayyan ID	Reference	Primary reason
386283297	Effects of Selective Inhibitors of Serotonin Reuptake on the Anxiety Behavior and Activity of the Pituitary-Adrenal System in Prenatally Stressed Male Rats. NEUROCHEMICAL JOURNAL. (2011). DOI: 10.1134/S1819712410041038.	Wrong exposure
386283326	Chronic Prenatal Stress Affects Development and Behavioral Depression in Rats. STRESS-THE INTERNATIONAL JOURNAL ON THE BIOLOGY OF STRESS. (1998). DOI: 10.3109/10253899809167291.	Wrong exposure
386283372	Cross-fostering alleviates depression-like behavior mediated by EAAT2 and SNARE complex in prenatal stress offspring rat. PHARMACOLOGY BIOCHEMISTRY AND BEHAVIOR. (2021). DOI: 10.1016/j.pbb.2021.173269.	Wrong exposure
386283501	Hormonal and behavioral characteristics of prenatally stressed female rats in an experimental model of depression. Neuroscience and Behavioral Physiology. (2019). DOI: https://dx.doi.org/10.1007/s11055-019-00772-6.	Wrong exposure
386283539	Behavioral alterations of adolescent rats exposed to prenatal stress: neurobiological mechanisms and implications for psychiatric disorders. DOI: 10.1016/j.psyneuen.2023.106725.	Wrong exposure
386283545	Perinatal citalopram does not prevent the effect of prenatal stress on anxiety, depressive-like behaviour and serotonergic transmission in adult rat offspring. EUROPEAN JOURNAL OF NEUROSCIENCE. (2016). DOI: 10.1111/ejn.13150.	Wrong exposure
386283617	Epigenetic Modifications of GABAergic Interneurons Contribute to Deficits in Adult Hippocampus Neurogenesis and Depression-Like Behavior in Prenatally Stressed Mice. INTERNATIONAL JOURNAL OF NEUROPSYCHOPHARMACOLOGY. (2020). DOI: 10.1093/ijnp/pyaa020.	Wrong exposure
386283689	Grape seed proanthocyanidins improves depression-like behavior by alleviating oxidative stress and NLRP3 activation in the hippocampus of prenatally-stressed female offspring rats. JOURNAL OF HISTOTECHNOLOGY. (2021). DOI: 10.1080/01478885.2020.1861907.	Wrong exposure
386283703	Differential Effect of Prenatal Stress on the Expression of Corticotrophin-Releasing Hormone and its Receptors in the Hypothalamus and Amygdala in Male and Female Rats. JOURNAL OF NEUROENDOCRINOLOGY. (2011). DOI: 10.1111/j.1365-2826.2011.02117.x.	Wrong exposure
386283720	Ketamine role in the treatment of Maternal depression: effects on offspring behaviour. DOI: 10.21203/rs.3.rs-3969981/v1.	Wrong exposure
386283754	The effects of oxytocin on anxiety and depression in the prenatal stress context in 3-NP injected rats. PHYSIOLOGY AND PHARMACOLOGY. (2025). DOI: 10.61186/phypha.29.1.63.	Wrong exposure
386283761	Reduced Levels of NR1 and NR2A with Depression-Like Behavior in Different Brain Regions in Prenatally Stressed Juvenile Offspring. PLOS ONE. (2013). DOI: 10.1371/journal.pone.0081775.	Wrong exposure
386283763	Identification of short-acting κ -opioid receptor antagonists with anxiolytic-like activity. EUROPEAN JOURNAL OF PHARMACOLOGY. (2011). DOI: 10.1016/j.ejphar.2011.04.017.	Wrong exposure
386283776	Sexual differences in long-term effects of prenatal chronic mild stress on anxiety-like behavior and stress-induced regional glutamate receptor expression in rat offspring. INTERNATIONAL JOURNAL OF DEVELOPMENTAL NEUROSCIENCE. (2015). DOI: 10.1016/j.ijdevneu.2015.01.003.	Wrong exposure
386283789	Prenatal stress leads to deficits in brain development, mood related behaviors and gut microbiota in offspring. NEUROBIOLOGY OF STRESS. (2021). DOI: 10.1016/j.ynstr.2021.100333.	Wrong exposure
386283805	Prenatal stress: Effects on somatic and behavioral ontogeny in the rat. Dissertation Abstracts International. (1976).	Wrong exposure
386283838	Interplay of serotonin and kappa opioid systems in stress, anxiety, and depressive disorders. Dissertation Abstracts International: Section B: The Sciences and Engineering. (2023).	Wrong exposure

Rayyan ID	Reference	Primary reason
386283897	Genotype and prenatal and premating stress interact to affect adult behavior in rats. Science. (1965). DOI: https://dx.doi.org/10.1126/science.150.3705.1844 .	Wrong exposure
386283912	Involvement of NR1, NR2A different expression in brain regions in anxiety-like behavior of prenatally stressed offspring. BEHAVIOURAL BRAIN RESEARCH. (2013). DOI: 10.1016/j.bbr.2013.08.044.	Wrong exposure
386283927	Motor lateralization, behavioral despair and dopaminergic brain asymmetry after prenatal stress. PHARMACOLOGY BIOCHEMISTRY AND BEHAVIOR. (1997). DOI: 10.1016/S0091-3057(97)00009-9.	Wrong exposure
386283942	Prenatal stress alters the negative correlation between neuronal activation in limbic regions and behavioral responses in rats exposed to high and low anxiogenic environments. PSYCHONEUROENDOCRINOLOGY. (2007). DOI: 10.1016/j.psyneuen.2007.03.013.	Wrong exposure
386283946	The impact of prenatal stress on insulin-like growth factor-1 and pro-inflammatory cytokine expression in the brains of adult male rats: The possible role of suppressors of cytokine signaling proteins. JOURNAL OF NEUROIMMUNOLOGY. (2014). DOI: 10.1016/j.jneuroim.2014.08.001.	Wrong exposure
386283981	The effect of prenatal stress on rat offspring anxiety. ZHURNAL VYSSHEI NERVNOI DEYATELNOSTI IMENI I P PAVLOVA. (1996).	Wrong exposure
386284007	Female Early Adult Depression Results in Detrimental Impacts on the Behavioral Performance and Brain Development in Offspring. CNS NEUROSCIENCE & THERAPEUTICS. (2012). DOI: 10.1111/j.1755-5949.2012.00324.x.	Wrong exposure
386284014	Gestational stress: Effects on open-field behaviour and heart rate reactivity in rat offspring. BIOLOGICAL PSYCHOLOGY. (1975). DOI: https://dx.doi.org/10.1016/0301-0511%2875%2990037-X .	Wrong exposure
386284036	Prenatal stress is a vulnerability factor for altered morphology and biological activity of microglia cells. FRONTIERS IN CELLULAR NEUROSCIENCE. (2015). DOI: 10.3389/fncel.2015.00082.	Wrong exposure
386284089	Maternal Mirtazapine Selectively Enhances Hippocampal LTP without Reversing Stress-Associated Alterations in Basal Transmission, Short-Term Plasticity, or Open Field Behaviour in Rat Offspring. DRUG RESEARCH. (2025). DOI: 10.1055/a-2682-5112.	Wrong exposure
386284128	Prenatal stress leads to chromatin and synaptic remodeling and excessive alcohol intake comorbid with anxiety-like behaviors in adult offspring. NEUROPHARMACOLOGY. (2018). DOI: 10.1016/j.neuropharm.2018.07.010.	Wrong exposure
386284145	Hippocampal metabolic alteration in rat exhibited susceptibility to prenatal stress. JOURNAL OF AFFECTIVE DISORDERS. (2019). DOI: 10.1016/j.jad.2019.08.002.	Wrong exposure
386284153	Maternal stress programs a demasculinization of glutamatergic transmission in stress-related brain regions of aged rats. GEROSCIENCE. (2022). DOI: 10.1007/s11357-021-00375-5.	Wrong exposure
386284157	Effect of Valeriana fauriei extract on the offspring of adult rats exposed to prenatal stress. INTERNATIONAL JOURNAL OF MOLECULAR MEDICINE. (2016). DOI: 10.3892/ijmm.2016.2589.	Wrong exposure
386284195	H3K9 Acetylation of Tph2 Involved in Depression-like Behavior in Male, but not Female, Juvenile Offspring Rat Induced by Prenatal Stress. NEUROSCIENCE. (2018). DOI: 10.1016/j.neuroscience.2018.03.043.	Wrong exposure
386281952	A study on behavioural effects of laboratory rats (albino wistar) after the sub-chronic noise stress. International Journal of Research in Medical. (2018).	Wrong outcome

Rayyan ID	Reference	Primary reason
386281970	Effects of perinatal stress on the emotion-related behavior of the adolescent mouse. BIOLOGICAL & PHARMACEUTICAL BULLETIN. (2006). DOI: 10.1248/bpb.29.1036.	Wrong outcome
386282738	The Effects of Acoustic White Noise on the Rat Central Auditory System During the Fetal and Critical Neonatal Periods: A Stereological Study. Noise & health. (2017). DOI: 10.4103/1463-1741.199239.	Wrong outcome
386283561	[NOISE-STIMULATION EFFECTS ON PREGNANT RATS AND ITS FETUSES]. Nihon Jibiinkoka Gakkai kaiho. (1964). DOI: 10.3950/jibiinkoka.67.6_843.	Wrong outcome
386283068	Long-term recovery from hippocampal-related behavioral and biochemical abnormalities induced by noise exposure during brain development. Evaluation of auditory International Journal of. (2014). DOI: 10.1016/j.ijdevneu.2014.06.002.	Wrong population
386283208	Prenatal noise stress aggravates cognitive decline and the onset and progression of beta amyloid pathology in a mouse model of Alzheimer's disease. NEUROBIOLOGY OF AGING. (2019). DOI: 10.1016/j.neurobiolaging.2019.01.019.	Wrong population
386283398	Prenatal effects of buspirone and stress on behavioral reactions in rat pups of different sexes during period of ontogeny with low level of brain serotonin. DOI: 10.17816/ped5190-96.	Wrong population
386283985	Chronic traffic noise stress accelerates brain impairment and cognitive decline in mice. EXPERIMENTAL NEUROLOGY. (2018). DOI: 10.1016/j.expneurol.2018.06.011.	Wrong population
386282791	Gender differences in the effects of prenatal stress on brain development and behaviour. NEUROCHEMICAL RESEARCH. (2007). DOI: 10.1007/s11064-007-9339-4.	Wrong publication type
386283438	Sex Differences in the Behavioral and Metabolic Effects of Prenatal Stress in Early but not Late Adulthood in Mice. DOI: 10.1124/jpet.513.946110.	Wrong publication type
386284125	[Comparative efficacy of selective serotonin reuptake inhibitors in young prenatally stressed female rats]. Eksperimental'naia i klinicheskaia farmakologiya. (2010).	Wrong publication type
386284175	4.1 Can Maternal N-Acetyl Cysteine Rescue the Effects of Prenatal Stress on Behavior and the Brain?. DOI: 10.1016/j.jaac.2017.09.217.	Wrong publication type

Appendix B. Full-text report not retrieved (n = 1)

This retrieval outcome is not included among the 74 assessed full-text exclusions.

Rayyan ID	Reference	Retrieval outcome
386283651	Impact of Maternal Road Traffic Noise Exposure on Hippocampal Spatial Memory and Neurogenesis of Sprague Dawley Offspring: Therapeutic Effects of Edaravone, Melatonin, and Enriched Environments. Central nervous system agents in medicinal chemistry. (2025). DOI: 10.2174/0118715249381371251019170111.	Full text not available

Appendix C. Additional included reports identified outside the formal strategy (n = 4)

These reports were added to the final included-study set after general/benchmark searching or study linkage and were not counted among the 87 Rayyan full-text records.

Study ID	Reference	Identification route
abramova_2021_front	The Behavior and Postnatal Development in Infant and Juvenile Rats After Ultrasound-Induced Chronic Prenatal Stress. DOI: 10.3389/fphys.2021.659366.	Benchmark/general search
arjunan_2023_stress	Impacts of Maternal Noise Exposure on Risk of Stillbirth and Oxidative Stress-Induced Neurobehavioral Changes in Offspring. DOI: 10.3390/stresses3030037.	Benchmark/general search
oliveira_2015_jbehavbraisci	Intrauterine Exposure to Chronic 22 kHz Sound Affects Inhibitory Avoidance and Serotonergic Parameters in Forebrain Areas of Dams and Rat Offspring. DOI: 10.4236/jbbs.2015.52003.	Benchmark/general search
abramova_2023_biopsy	Blood BDNF and neuropeptide concentrations as potential biomarkers of behavioral impairments in rats with prenatal ultrasound stress experience.	Related report identified during study linkage

Identification and selection of prenatal chronic-noise studies

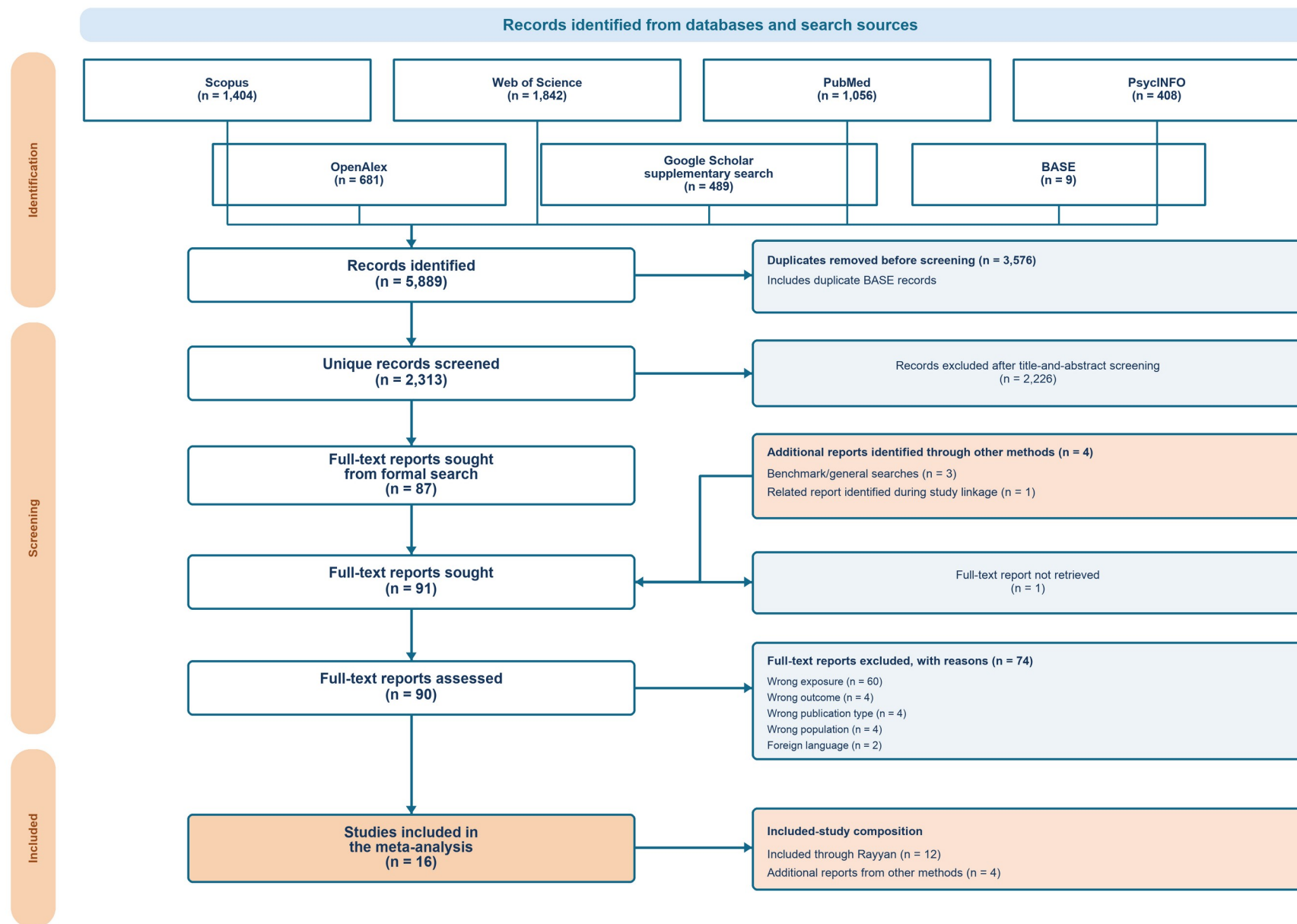


Figure 1. PRISMA-style flow diagram for identification, screening, full-text assessment, and inclusion in the meta-analysis.